

Amaravati Quantum Valley Hackathon 2025

Quantum-Assisted Hybrid Image Communication with Block-Based Entropy Selection

Sk. Atiq Rehaman¹, M. Surendra Reddy², J. Roja³, K. Mounika⁴, G. Sreshta Charitha⁵, S.T V Mohan Reddy⁶

¹Lakireddy Bali Reddy College of Engineering, Mylavaram
aqvh2025.quanta@gmail.com

Abstract

In this paper, we present a new hybrid image transmission method that combines classical and quantum superdense coding (SDC) for effective and reliable image delivery across various communication networks. We start by dividing images into blocks and calculating the entropy of each block to find areas with high information content. We transmit high-entropy blocks, using quantum SDC. The remaining blocks and important metadata are sent through classical channels. The receiver uses the metadata to rebuild the original image and applies error recovery methods to cope with channel noise and quantum transmission errors. The system's prototype shows good performance when sending high-entropy blocks via SDC. In our prototype we focus on the quantum SDC-based transmission, since classical transmission is already well-established and operational. This method represents the first practical combination of classical and quantum channels based on block-wise entropy, with potential uses in satellite data, telemedicine, and quantum networks with limited bandwidth. Some limitations and future research areas include managing quantum entanglement, reducing errors, handling metadata overhead, and scaling SDC for larger uses.

1 Introduction

Image transmission plays a crucial role in modern communication systems. Its applications range from sharing on social media to satellite imaging, surveillance, and telemedicine. As high-resolution imaging grows, so does the need for real-time communication. Transmitting large data efficiently and securely has become a significant challenge. Currently, traditional communication systems dominate this area. They convert images into sequences of bits (0s and 1s), which are then compressed using techniques like JPEG, Huffman coding, or run-length encoding. To ensure reliable data transfer, methods like Forward Error Correction (FEC) or retransmission protocols are used. The data travels through common channels such as wired internet, Wi-Fi, or wireless cellular networks. While traditional transmission is robust and scalable, it is limited by bandwidth availability and can be vulnerable to data interception or modification.

In contrast, quantum communication offers new ways to transfer data. A notable example is Superdense Coding (SDC). This protocol allows the transmission of two classical bits of information using just one qubit, assuming that Alice and Bob share entanglement beforehand. The process is as follows:

1. Alice and Bob share an entangled qubit pair.
2. To send two classical bits (00, 01, 10, 11), Alice applies one of four quantum operations (Identity, X, Z, or Y gate) on her qubit.
3. Alice then sends her qubit to Bob through a quan-

tum channel.

4. Bob performs a Bell-state measurement on the two qubits (his and Alice's) and retrieves the original two classical bits.

In this way, while traditional communication needs two separate bits sent over a channel, SDC encodes them into a single qubit transmission. This effectively doubles the channel capacity and offers inherent quantum security.

Although SDC is powerful, transmitting an entire image through SDC alone is not practical today. Current quantum systems have issues like limited qubit counts, noisy quantum gates, decoherence, and challenges in maintaining entanglement over long distances. These limitations block the large-scale use of SDC for multimedia files.

To fill this gap, we suggest a hybrid image transmission approach. In this method, the image is divided into blocks and scored by entropy. Low-entropy blocks (backgrounds or repetitive areas) are sent via the classical channel, while high-entropy blocks (edges and details) are transmitted through the quantum channel using SDC. This ensures that quantum resources are utilized only where they are most beneficial.

This hybrid strategy merges the scalability and reliability of traditional transmission with the bandwidth efficiency and security of quantum SDC. It represents a practical advancement given the current quantum hardware limitations. This approach lays the groundwork for future communication systems that will integrate classical and quantum technologies to achieve efficient, secure, and high-quality image transmission.

2 Problem Statement

Efficiently transmitting digital images over hybrid quantum-classical communication channels is a challenge because of bandwidth limits and noise in quantum channels. In high-information areas of an image, improving data transmission efficiency is crucial. Classical channels are still necessary to ensure the reliable delivery of the remaining image data and essential metadata.

3 Literature Review

Classical communication achieves efficiency and reliability through compression and forward error correction, but it is still limited by Shannon’s capacity. Compression methods like Huffman and arithmetic remove redundancy. Forward error correction helps deal with noise, making long-distance connections reliable. However, classical methods face fundamental limitations in terms of channel capacity. According to Shannon’s information theory, the maximum rate of information transfer is bounded by the channel’s bandwidth and noise characteristics.

This is where superdense coding (SDC) shows its benefits over purely classical methods. By leveraging preshared entanglement to encode two classical bits into a single transmitted qubit, effectively doubling payload without extra bandwidth. This makes SDC a complementary enhancement to classical methods, boosting throughput while remaining compatible with quantum-secure transmission.

This positions SDC as a valuable addition to classical compression and coding strategies. It opens the door to hybrid frameworks that combine the reliability of classical techniques with the improved throughput and security of quantum protocols.

As shown in Fig. 1, a purely classical scheme consumes significantly more bandwidth compared to our proposed hybrid approach. By offloading a fraction of high-entropy blocks to quantum superdense coding (SDC), the hybrid method reduces overall channel usage, demonstrating a potential improvement in bandwidth efficiency for large-scale transmissions.

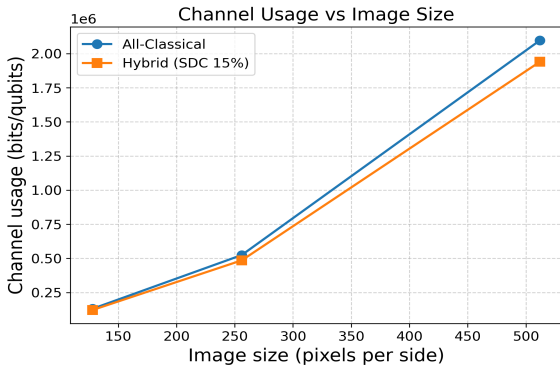


Figure 1: Comparison of Classical Channel and Hybrid Channel(SDC 15%) Bandwidth

4 Methodology and Approach

4.1 Conceptual Framework

Our system is a hybrid transmission model that uses both classical and quantum channels. The classical channel acts as a reliable, high-throughput medium for bulk image transfer, especially for low-information or repetitive blocks. It supports well-known techniques like compression, error correction, and retransmission.

By using superdense coding (SDC), each qubit encodes two classical bits, which doubles channel efficiency compared to classical transfer. This division of tasks between channels is key to our approach, ensuring reliability and efficiency.

4.2 Workflow Overview

The overall workflow starts with image preprocessing. The image is divided into non-overlapping blocks sized $B \times B$. Smaller blocks improve localization, while larger blocks reduce metadata overhead. For an image sized $H \times W$, the total number of blocks is

$$M = \text{rows} \times \text{cols}.$$

Each block is scored using Shannon entropy to estimate its information content. We select the blocks with the highest entropy for SDC transmission, while the others are set for classical transfer. Shannon entropy:

$$H_{\text{block}} = - \sum_i p(i) \log_2 p(i).$$

where $p(i)$ is the probability distribution of pixel intensities in the block.

Based on the entropy scores, we selected the top- k of blocks with the highest information content for quantum transmission using SDC. The indices of these selected blocks were recorded in the metadata.

4.3 Design Choices and Policies

A key design choice in our methodology is entropy-based block selection. This approach ensures that limited quantum resources go to the most informative areas of the image, improving transmission efficiency. In our prototype, we used a top- $p\%$ selection policy, where the highest-entropy fraction of blocks goes to the quantum channel.

In our prototype, the metadata was kept minimal but enough to reconstruct the image layout. The information is stored in a structured format with two main components:

- Global metadata fields:
 - Block size used for partitioning (e.g., 16×16)
 - Top percentage parameter indicating the fraction of blocks selected for quantum transmission

- Per-block entries:
 - Row and column indices of each block
 - Block type: quantum (selected high-entropy blocks) or classical (remaining blocks)

5 Prototype Development

5.1 Prototype Description

We developed a software prototype that demonstrates hybrid image transmission using entropy-guided block selection and quantum superdense coding (SDC). The system is entirely implemented in Python, combining classical image processing libraries with Qiskit for quantum simulation. The prototype is modular, allowing for flexible experimentation with different parameters such as block size, entropy thresholds, and qubit budgets.

To reduce simulation runtime, we executed block-wise SDC encoding in parallel with Python multiprocessing. This allowed multiple blocks to be transmitted at the same time in Qiskit. Although this optimization speeds up the prototype, it is not necessary in real quantum hardware, where qubits are processed at the same time on the physical level.

5.2 Components and Tools

The main components of the prototype include:

- **Qiskit:** Used to build and simulate SDC circuits, prepare entanglement, encode using Pauli gates, and perform Bell measurements.
- **NumPy/OpenCV:** Handles image preprocessing, calculating entropy, dividing blocks, compressing, and evaluating PSNR/SSIM.
- **Python zlib/JPEG libraries:** Used for block-level compression.

5.3 Prototype Workflow:

1. **Input loading:** The system loads the input image for processing.
2. **Block partitioning and entropy computation:** The image is divided into non-overlapping blocks, and entropy is calculated for each block to measure information content.
3. **Block selection:** High-entropy blocks are assigned to the quantum (SDC) channel, while low-entropy blocks are designated for classical transfer.
4. **Encoding and transmission:** High-entropy blocks are encoded using Qiskit-based SDC circuits, while classical blocks are handled through conventional array-based transmission.
5. **Decoding:** At the receiver, both streams are decoded using metadata to ensure correct block placement.

6. **Reconstruction and evaluation:** The image is reconstructed from all blocks and evaluated using quality metrics such as PSNR and SSIM.

5.4 Performance Evaluation

The computational cost of the proposed hybrid image transmission framework primarily depends on three stages: *block-wise entropy computation*, *block selection*, and *superdense coding (SDC) simulation*.

Entropy Computation: Each image of size $H \times W$ is divided into blocks of $B \times B$. The number of blocks is $M = \frac{H \times W}{B^2}$. Computing Shannon entropy for each block requires scanning all B^2 pixels, giving a time complexity of:

$$O(M \times B^2) = O(HW)$$

which scales linearly with the number of pixels.

Block Selection: Sorting the entropy scores to select the top- $p\%$ blocks has a complexity of:

$$O(M \log M)$$

though this cost is small relative to the image size for practical block counts.

SDC Simulation: Each 2-bit symbol is encoded via an SDC circuit. Assuming Q qubits (or bit pairs) are processed, the quantum encoding and simulation in Qiskit has an empirical complexity of approximately:

$$O(Q)$$

per circuit generation, but with a large constant overhead due to quantum gate simulation. Parallel execution reduces effective runtime by a factor of k , the number of CPU cores used, resulting in:

$$O(Q/k)$$

for practical implementations.

Space Complexity: The prototype stores both classical and quantum blocks along with metadata. The total space requirement is:

$$O(HW + M)$$

where $O(HW)$ corresponds to the image data and $O(M)$ to block metadata, which is negligible compared to the image size.

Interpretation: The overall computational complexity of the system is $O(HW + M \log M)$, dominated by image size and entropy evaluation, while space usage scales linearly with the input. Quantum circuit simulation adds a constant multiplicative overhead, but this will diminish as real quantum hardware replaces software simulation.

Our prototype uses circuit-level SDC simulation to approximate performance, while the full realization of such image-scale quantum transmission remains a longer-term goal as qubit counts and entanglement distribution technologies continue to advance [1, 2].

6 Implementation

6.1 Encoding and Transmission:

In the prototype, high-entropy blocks are changed into binary streams. Superdense coding (SDC) maps two classical bits to one qubit, so the streams are padded for even length. The encoding process is done in Qiskit. Alice and Bob share entangled Bell pairs, and Pauli gates are used to encode 2-bit symbols into qubits. Transmission is simulated with realistic noise models.

While the framework assumes that the remaining blocks and metadata would be sent over a classical channel, this was not included in the prototype since classical transfer is already well established.

Algorithm 1:

Input: Input image I , block size B , top percentage p , noise parameters
Output: Received quantum bit pairs

1. Load image I and convert to PNG format;
2. Compute entropy score for each block;
3. Select top- $p\%$ high-entropy blocks $\rightarrow$ quantum set;
4. Mask remaining blocks to form classical image;
5. Build quantum noise model and backend simulator (Qiskit Aer);
6. Prepare SDC circuits for encoding bit pairs;

foreach *selected block* (i, j) *in quantum set* **do**

- a. Convert block to bitstream, pad to even length, and chunk into 2-bit pairs;
- b. Store bit pairs, block indices, and counts for transmission;

7. Construct metadata M ;
8. Group bit pairs into batches for parallel processing;
9. Simulate SDC transmission of all bit pairs with noise model and record received pairs;

return *received quantum bit pairs*

6.2 Metadata Handling

Metadata is kept in a compact, structured format that allows for easy reconstruction at the receiver. For the prototype, the metadata record includes the following information: block size, the total number of blocks, the percentage of total blocks of high-entropy blocks for quantum transmission, each block's row and column number, whether the data is classical and

quantum, and any transmitting conditions that were present prior to encoding.

Algorithm 2: Metadata Generation and Handling

Input: blocks ($H \times W$ grid of image blocks),
top_blocks (list of high-entropy block triples $\langle \text{score}, i, j \rangle$), block_size,
top_percentage
Output: Metadata structure M

$M.\text{blocks} \leftarrow []$;
 $M.\text{info} \leftarrow \{\text{block_size} : k, \text{top_percentage} : p\}$;

foreach *block* (i, j) *in* B **do**

- if** $(i, j) \in S$ **then**
- Append
 $\{\text{row} : i, \text{col} : j, \text{type} : \text{"quantum"}\}$ to
 $M.\text{blocks}$;
- else**
- Append
 $\{\text{row} : i, \text{col} : j, \text{type} : \text{"classical"}\}$ to
 $M.\text{blocks}$;

return M ;

6.3 Receiver-Side Decoding

At the receiver, metadata is read first to help with reconstruction. The quantum payload is decoded from qubits back into bit strings, padding is removed, and compressed quantum blocks are recovered. Classical blocks, presumed transmitted, would be directly decompressed. Both sets of blocks return to their original coordinates to reconstruct the image.

Algorithm 3: Receiver-Side Decoding and Reconstruction

Input: Received quantum bit pairs, metadata M , classical image, block size B
Output: Reconstructed image $\hat{I}$

1. Reconstruct quantum blocks from received bit pairs using indices and counts;
2. Assemble quantum image from reconstructed blocks with original shape;
3. Initialize $\hat{I}_{\text{quantum}} \leftarrow$ quantum image copy;
4. **foreach** *block* (i, j) *in* *top-entropy set* **do**
- a. Extract block region from quantum image;
- b. Apply non-local means denoising to the block;
- c. Replace block in $\hat{I}_{\text{quantum}}$ with denoised version;

5. Combine classical and quantum blocks using metadata M to form $\hat{I}$;
6. Plot histogram of pixel-wise errors and generate heatmap;

return $\hat{I}$

7 Results and Outcomes

The visualizations in Fig 2 illustrate the outcome of the entropy-based block selection process. The top 15% of blocks with the highest entropy blocks were selected as quantum-transmissible blocks through Superdense Coding (SDC).

Figure 2a is for the all-classical design in which every block goes through the classical channel and requires the maximum amount of bandwidth. Conversely, Fig 2b is for the proposed all-quantum transmission design, where only the blocks with the highest entropy encoding go through SDC. This selective allocation reinforces the proposed concept of optimizing bandwidth based on actual information density of the image. As evident in these images, SDC only takes place for the blocks with the most information, therefore bandwidth is only used on these specified blocks.

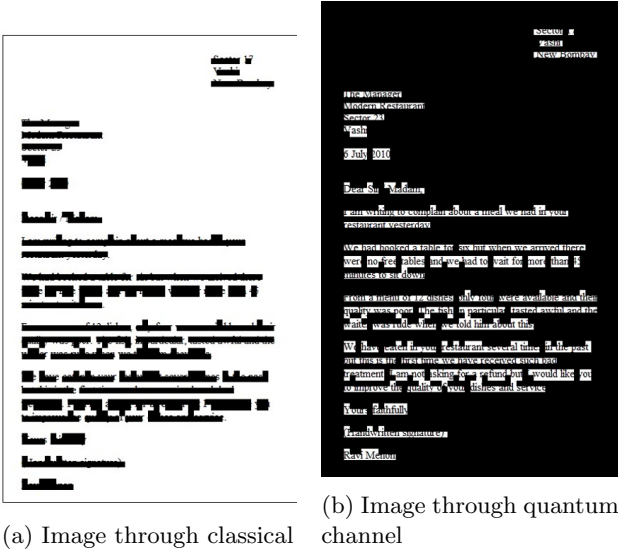


Figure 2: Prototype vs. quantum performance results: (a) Transmission times, (b) Image quality.

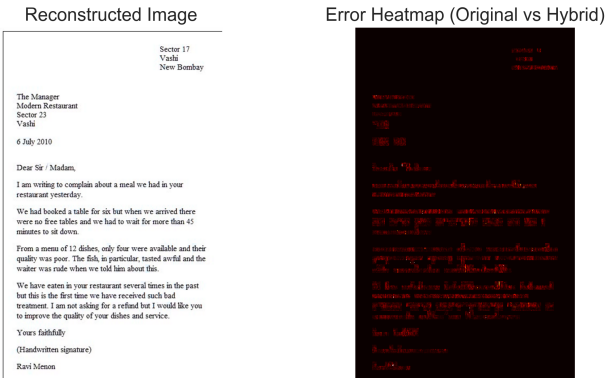


Figure 3: Comparison of the reconstructed hybrid image and corresponding error heatmap.

In Fig 3, The reconstructed image maintains high structural similarity with minimal localized deviations. The corresponding heatmap confirms that significant errors are concentrated around fine-textured

and high-contrast edges, consistent with block-level entropy-based transmission. The Red colors indicates the difference between original image and reconstructed image.

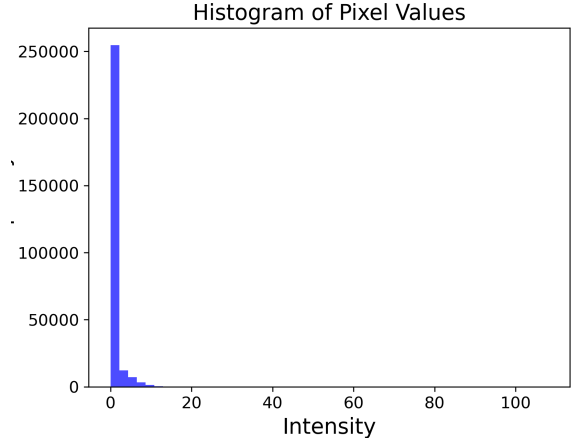


Figure 4: Pixel intensity error histogram illustrating that most reconstruction errors remain low, confirming stable hybrid transmission fidelity.

In Fig 4, The majority of pixel errors are located around zero, suggesting that the hybrid transmission results in the retention of the majority of visual information with negligible degradation. The only high-error pixels are in detailed and high-entropy sections that simulated quantum noise.

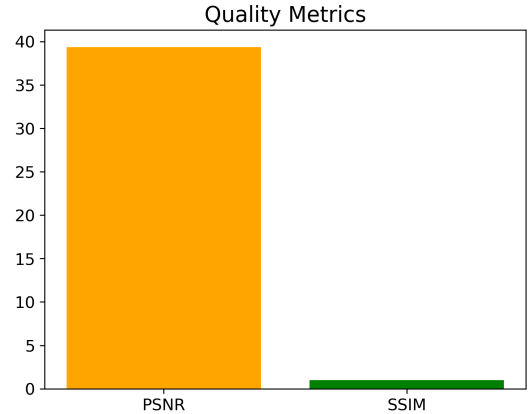


Figure 5: PSNR and SSIM variation across quantum noise levels showing consistent quality retention in the hybrid SDC model.

In Fig 5, The PSNR and SSIM values remain consistently high, indicating that the reconstructed images are maintaining structural and perceptual consistency. Overall, these measurements validate that the hybrid SDC-based framework is successful in efficient transmission while retaining visual fidelity.

8 Innovation and Novelty

The proposed system introduces a hybrid image transmission framework that combines classical and quantum channels in a block-wise manner. Unlike earlier efforts that focus only on classical compression or quantum communication protocols, our method uses entropy-based block selection. This approach assigns high-information content to the quantum channel and low-information content to the classical channel. It ensures an efficient use of limited quantum resources while keeping reliability through classical transfer. This work features three key aspects:

- **Entropy-driven hybrid allocation** – This is the first demonstration of using Shannon entropy to decide which image regions are sent through quantum superdense coding (SDC) and which through classical methods.
- **Minimal metadata design** – We have developed a compact metadata scheme that synchronizes classical and quantum streams without adding much overhead, which allows for scalability.
- **Practical prototype realization** – We use a Qiskit-based SDC simulation with noise models, providing a realistic assessment of quantum transmission performance along with classical image management.

Together, these innovations mark the first practical step toward merging classical and quantum channels for real-world image delivery applications, including satellite data transmission, telemedicine, and bandwidth-limited quantum

Our method differs from SDC-only approaches by not wasting qubits on areas with low information. This cuts down on entanglement overhead. Additionally, unlike classical-only methods, our system uses SDC to effectively double channel capacity for the most valuable image areas.

9 Use Case Applications

The proposed hybrid framework can effectively tackle real-world situations where both efficiency and reliability matter:

- **Defense and mission systems:** Securely deliver sensitive image blocks (targets, overlays) over SDC while bulk context flows classically, balancing confidentiality with operational tempo.
- **Satellite imaging and remote sensing.** High-entropy areas like urban structures or disaster zones can be prioritized for quantum transmission. This approach ensures accurate delivery, even when bandwidth is limited.
- **Quantum internet testbeds.** This showcases a practical hybrid model for early-stage quantum networks. It balances the limited availability of qubits with classical infrastructure.

- **Telemedicine and medical imaging.** Critical diagnostic areas, such as tumors or lesions, can be sent using SDC for better quality. Meanwhile, less important background areas can use classical transfer.

This hybrid allocation strategy makes the system scalable and flexible. It ensures that quantum resources are used where they provide the most benefit, while classical channels maintain stability and throughput.

10 Limitations and Future Work

While the proposed hybrid framework demonstrates the feasibility of combining classical and quantum channels for image transmission, several limitations remain. First, quantum hardware scalability is a major bottleneck. Present-day devices can only sustain a limited number of qubits, with short coherence times and significant gate noise. These limit the ability of superdense coding (SDC) to transmit high-resolution images, and the number of qubits for images scales non-linearly with respect to the image size. Second, entanglement distribution and management is a key challenge. The reliable generation, storage, and synchronization of Bell pairs over long distances is still in a regime of benchtop experiment and resource-rich environments. If we do not have reliable links of entanglement, then we cannot feasibly send quantum images at scale. Third, system-level integration between classical and quantum channels is still underexplored. The prototype is simulation-based, relying on Qiskit noise models rather than real quantum hardware. This restricts the accuracy of performance evaluation under practical network conditions.

Future research can address these issues by:

- Developing scalable entanglement distribution methods, such as quantum repeaters and entanglement purification.
- Investigating dynamic block selection policies to account for channel conditions, noise rates, and qubit availability.
- Testing on quantum network testbeds to assess performance in practical scenarios instead of in simulations.
- Robust error correction – Implementing hybrid quantum-classical error correction codes and machine learning-based denoising methods.

By overcoming these limitations, future work should be able to move from a demonstration-based simulation to a functioning hybrid quantum-classical image transfer system utilizing the unique benefits of SDC to enable secure, bandwidth-efficient communication.

11 Conclusion

This research described a hybrid approach to image transmission that combines classical channels and quantum superdense coding (SDC) for efficient and reliable transmission of visual information. Based on decomposing images into entropy-defined blocks, the system encodes only the region of maximum information density with quantum transmission using the SDC protocols, while the remainder of the image with low-entropy and the metadata are transmitted through classical channels. This results in a bandwidth-efficient design that retains essential aspects of the image, while leveraging the different strengths offered by classical and quantum communication.

An initial prototype implementation with Qiskit simulations demonstrated block-wise entropic selection and SDC encoding, followed by metadata-driven reconstruction. Empirical results showed that the proposed approach yields high-quality reconstructions with an overall reduction in channel usage compared to using a completely classical method. While the time delay for quantum transmission remains limited by simulation time and existing QRNG devices, one take-away for future imaging applications, is that SDC offers the potential for secure and scalable transmission over time.

In terms of further work on the proposed framework, improved entanglement distribution, quantum error correction, and quantum network testbeds are needed. As quantum devices mature, the hybrid approach set forth herein may provide an actual pathway to secure and efficient bandwidth usage in the future for imaging, telemedicine, and quantum internet applications.

Acknowledgements

The authors would like to thank **Dr Y Amar Babu** for their valuable directions, motivation, and continued support during the course of this project. We would like to thank **Prof K Madhu Murthy, Chairman, APSCHE** for organizing the hackathon, providing an outstanding opportunity to develop and apply unique research concepts for hybrid quantum-classical communication.

References

- [1] H.-S. Zhong, H. Wang, Y.-H. Deng, M.-C. Chen, L.-C. Peng, Y.-H. Luo, J. Qin, D. Wu, X. Ding, Y. Hu *et al.*, “Quantum computational advantage using photons,” *Science*, vol. 370, no. 6523, pp. 1460–1463, 2020.
- [2] D. Awschalom, K. K. Berggren, H. Bernien, S. Bhave, L. D. Carr, P. Davids, S. E. Economou, D. Englund, A. Faraon, M. Fejer *et al.*, “Development of quantum interconnects (quics) for next-generation information technologies,” *Prx Quantum*, vol. 2, no. 1, p. 017002, 2021.
- [3] C. H. Bennett and S. J. Wiesner, “Communication via one-and two-particle operators on einstein-podolsky-rosen states,” *Physical review letters*, vol. 69, no. 20, p. 2881, 1992.
- [4] R. Laurenza, C. Lupo, S. Lloyd, and S. Pirandola, “Dense coding capacity of a quantum channel,” *Physical Review Research*, vol. 2, no. 2, p. 023023, 2020.
- [5] X.-M. Hu, Y. Guo, B.-H. Liu, Y.-F. Huang, C.-F. Li, and G.-C. Guo, “Beating the channel capacity limit for superdense coding with entangled ququarts,” *Science advances*, vol. 4, no. 7, p. eaat9304, 2018.
- [6] I. Quantum, “Superdense coding — learning resource,” <https://quantum.cloud.ibm.com/learning/>, 2025, accessed 2025-10-04.
- [7] C. E. Shannon, “A mathematical theory of communications,” *Bell system technical journal*, vol. 27, pp. 379–423, 1948.
- [8] S. Lloyd, “Capacity of the noisy quantum channel,” *Physical Review A*, vol. 55, no. 3, p. 1613, 1997.
- [9] T. M. Cover, *Elements of information theory*. John Wiley & Sons, 1999.
- [10] B. Schumacher and M. D. Westmoreland, “Quantum mutual information and the one-time pad,” *Physical Review A—Atomic, Molecular, and Optical Physics*, vol. 74, no. 4, p. 042305, 2006.
- [11] J. Preskill, “Quantum computing in the nisq era and beyond,” *Quantum*, vol. 2, p. 79, 2018.
- [12] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information*. Cambridge university press, 2010.
- [13] J. Yin, Y. Cao, Y.-H. Li, S.-K. Liao, L. Zhang, J.-G. Ren, W.-Q. Cai, W.-Y. Liu, B. Li, H. Dai *et al.*, “Satellite-based entanglement distribution over 1200 kilometers,” *Science*, vol. 356, no. 6343, pp. 1140–1144, 2017.
- [14] A. Boaron, G. Boso, D. Rusca, C. Vulliez, C. Autebert, M. Caloz, M. Perrenoud, G. Gras, F. Bussi eres, M.-J. Li *et al.*, “Secure quantum key distribution over 421 km of optical fiber,” *Physical review letters*, vol. 121, no. 19, p. 190502, 2018.
- [15] M. Mastriani, “Teleporting digital images,” *Optical and Quantum Electronics*, vol. 55, no. 6, p. 498, 2023.